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Project Title	Multi -Channel Marketing Attribution with Markov Chain Transitions
Week	Week 6

WEEK-6 TASK

Task Name

Multi -Channel Marketing Attribution with Markov Chain Transitions

Abstract

This project studies customer interactions with different marketing channels. It uses Markov Chain analysis to understand customer journeys and identify the contribution of each channel to final conversion.

Introduction

Customers usually interact with several marketing channels before making a purchase. Markov Chain helps represent these interactions as transitions and measures how each channel affects conversion.

Objectives

- To analyze 15 million user journey logs.
- To construct a transition probability matrix.
- To calculate conversion probabilities.
- To identify the most effective marketing channel.

Methodology

The user path data is collected, cleaned, and organized into channel sequences. Transition probabilities are calculated and used to construct the Markov Chain. Each channel is then removed to measure its effect on conversion.

Markov Chain & Transition Matrix

In the Markov Chain model, each marketing channel is treated as a state. The transition matrix represents the probability of moving from one channel to another, such as **Google → Instagram**.

Removal Effect

Removal Effect shows how much the conversion probability decreases when a particular marketing channel is removed. A higher value indicates greater channel contribution.

Formula

$$Removal\ Effect = \frac{Original\ Conversion - New\ Conversion}{Original\ Conversion}$$

Implementation / Algorithm

The system loads the user path logs and calculates channel transitions. It creates the transition matrix, calculates the original conversion probability, removes each channel one by one, and calculates the Removal Effect.

Sample Dataset

The sample dataset contains **User ID, Marketing Channel Path, and Conversion**.

Example:**U001:** Google → Instagram → Email → Conversion

Results

The Removal Effect values of all channels are compared. The channel with the highest value is identified as the most influential channel for conversion.

Conclusion

The project demonstrates how Markov Chain can be used for multi-channel marketing attribution. It helps businesses understand customer journeys and make better marketing decisions based on channel contribution.